

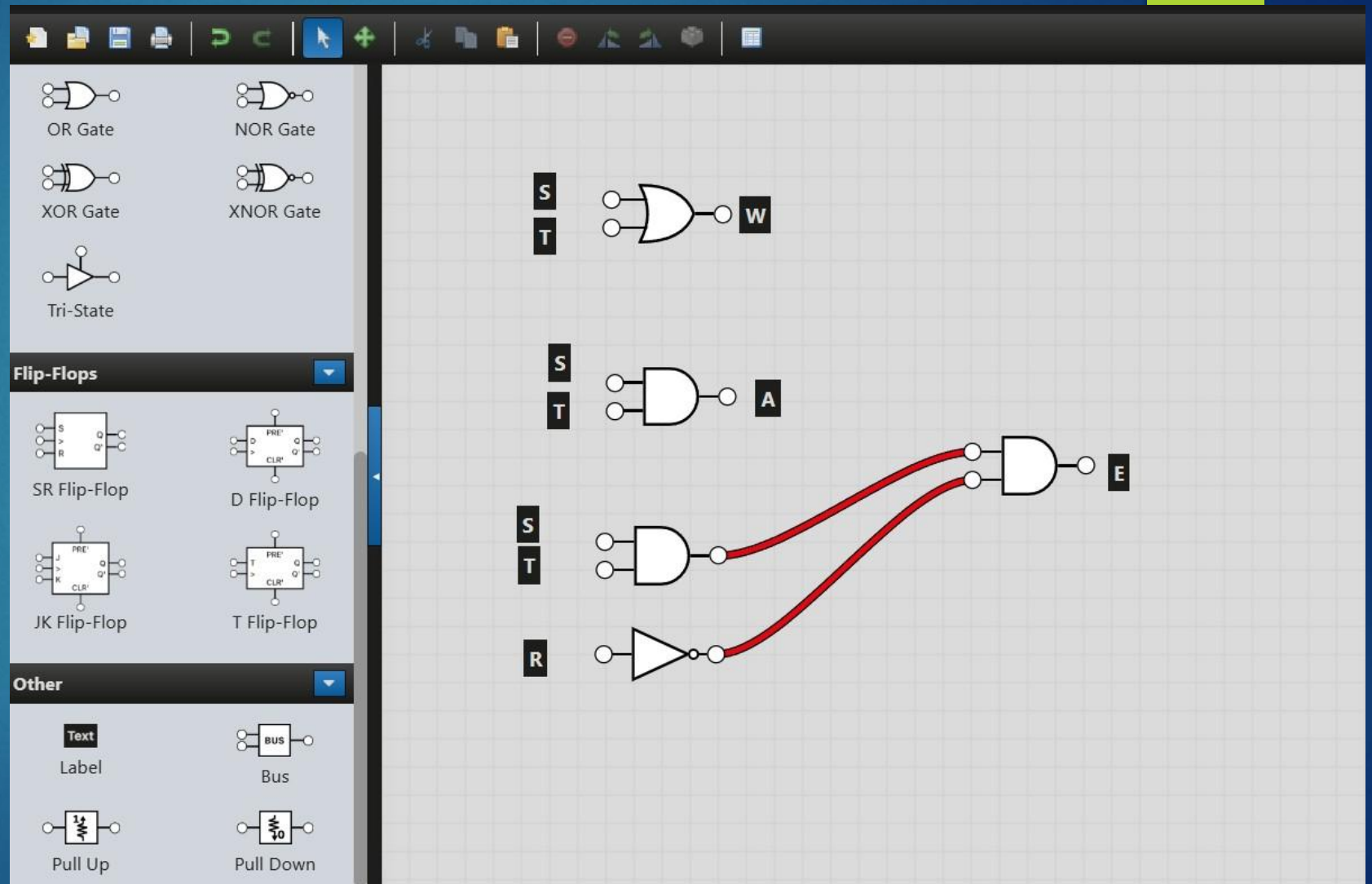
SMART FIRE / SMOKE WARNING SYSTEM

SUPPORTIVE DOCUMENT FILE



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LOGIC GATES DIAGRAM



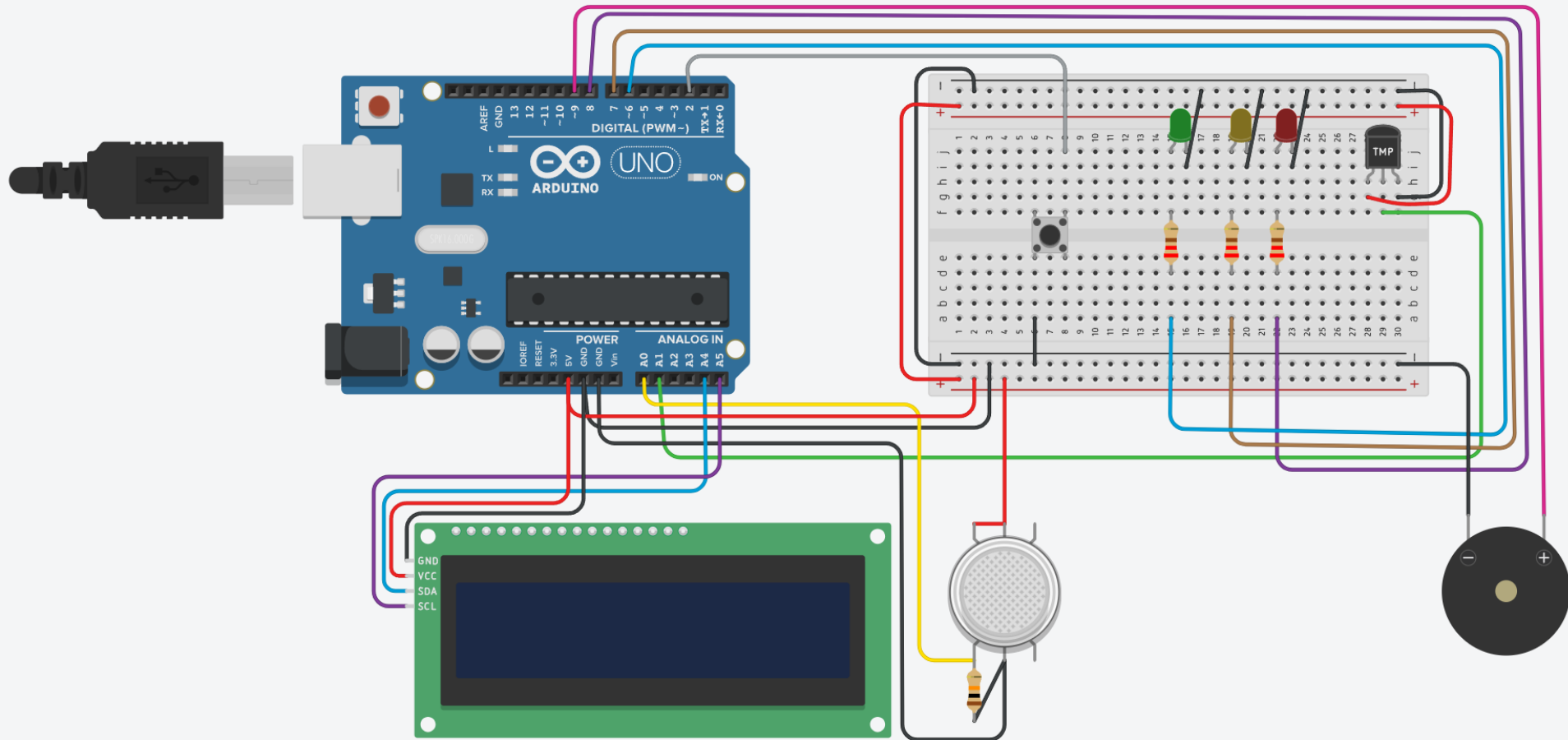
LOGIC.LY

TRUTH TABLE

W = OR A = AND E = (AND + NOT R)

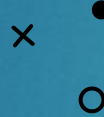
S	T	R	W	A	E
0	0	0	0	0	0
0	0	1	0	0	0
0	1	0	1	0	0
0	1	1	1	0	0
1	0	0	1	0	0
1	0	1	1	0	0
1	1	0	1	1	1
1	1	1	1	1	0

TINKERCAD DESIGN



https://www.tinkercad.com/things/4fJy21QMP8D-amazing-hillar/editel?returnTo=https%3A%2F%2Fwww.tinkercad.com%2Fdashboard&sharecode=hoFBjsEa2PsuOEFNwhZwsNTVHu7_IZqmhv1quBh3Kqw

TINKERCAD.COM



TINKERCAD

C++

```
Text 1 (Arduino Uno R3)

1 #include <Wire.h>
2 #include <LiquidCrystal_I2C.h>
3
4 LiquidCrystal_I2C lcd(0x27, 16, 2); // Tinkercad ALWAYS uses 0x
5
6 const int gasPin = A0;
7 const int tempPin = A1;
8 const int resetPin = 2;
9 const int greenLED = 6;
10 const int yellowLED = 7;
11 const int redLED = 8;
12 const int buzzer = 9;
13
14 unsigned long alarmStart = 0;
15 bool timerOn = false;
16 String lastState = "";
17
18 // --- LCD Helper ---
19 void setLCD(const char* line1, const char* line2) {
20   lcd.clear();
21   lcd.setCursor(0, 0);
22   lcd.print(line1);
23   lcd.setCursor(0, 1);
24   lcd.print(line2);
25 }
26
27 // --- All outputs OFF ---
28 void alloff() {
29   digitalWrite(greenLED, LOW);
30   digitalWrite(yellowLED, LOW);
31   digitalWrite(redLED, LOW);
32   digitalWrite(buzzer, LOW);
33 }
34
35 void setup() {
36   pinMode(greenLED, OUTPUT);
37   pinMode(yellowLED, OUTPUT);
38   pinMode(redLED, OUTPUT);
39   pinMode(buzzer, OUTPUT);
40   pinMode(resetPin, INPUT_PULLUP);
41
42   Serial.begin(9600);
43
44 }
```

Serial Monitor

```
SAFE
Gas: 487 Temp: 24.78
RESET
Gas: 487 Temp: 24.78
SAFE
Gas: 487 Temp: 24.78
SAFE
```

```
Text 1 (Arduino Uno R3)

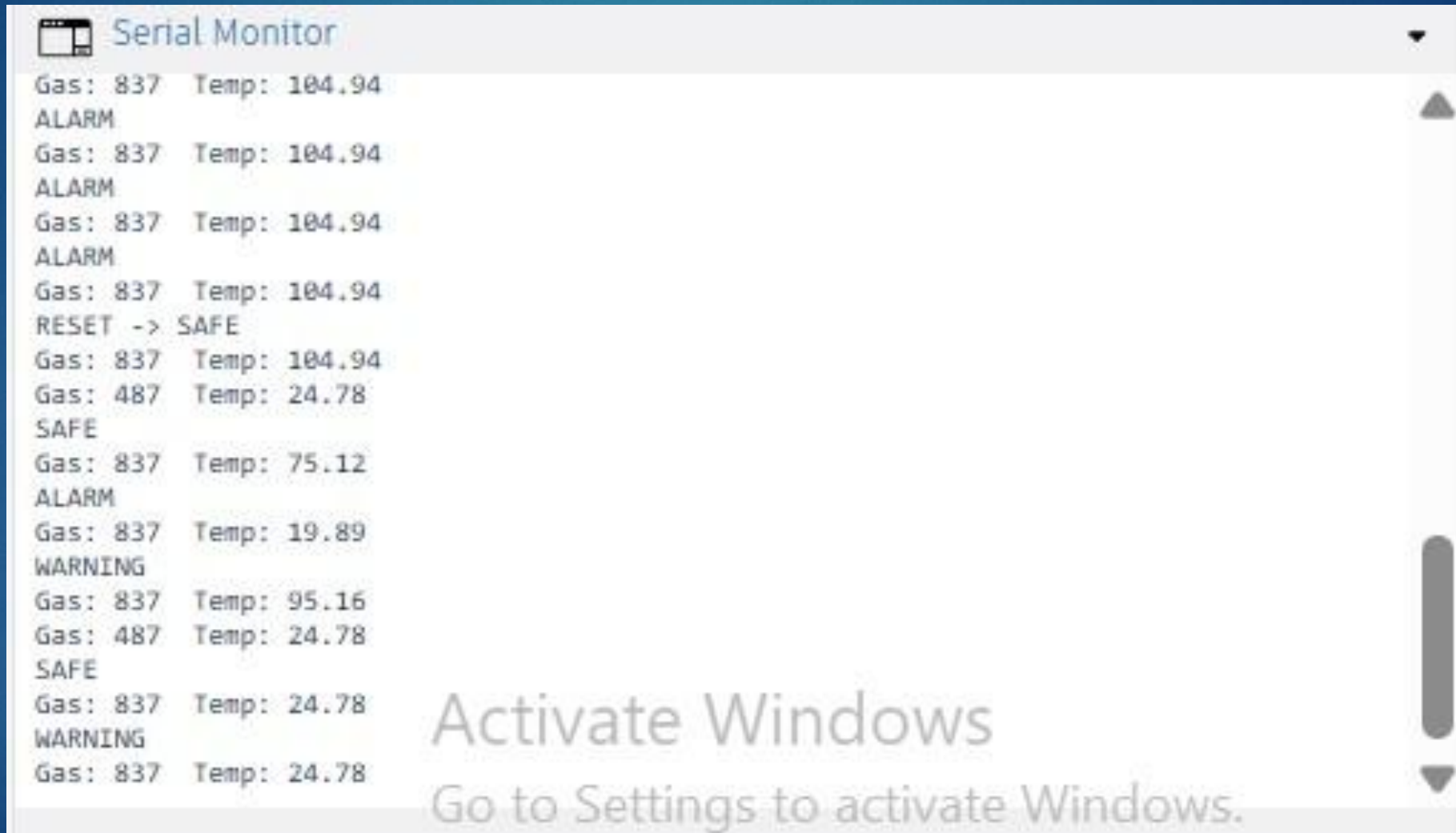
42 Serial.begin(9600);
43
44 // --- I2C + LCD INIT ---
45 Wire.begin();
46 delay(100); // Give LCD time to power up
47 lcd.init();
48 delay(50);
49 lcd.backlight();
50 delay(50);
51
52 setLCD(" Fire Detector ", " Starting... ");
53 delay(2000);
54
55 alloff();
56 digitalWrite(greenLED, HIGH);
57 setLCD(" SAFE ", " Monitoring... ");
58 lastState = "SAFE";
59 }
60
61 void loop() {
62   int gasVal = analogRead(gasPin);
63   float voltage = analogRead(tempPin) * (5.0 / 1023.0);
64   float tempC = (voltage - 0.5) * 100.0;
65   bool resetPressed = (digitalRead(resetPin) == LOW);
66
67   Serial.print("Gas: "); Serial.print(gasVal);
68   Serial.print(" Temp: "); Serial.println(tempC);
69
70   // --- RESET ---
71   if (resetPressed) {
72     alloff();
73     timerOn = false;
74     alarmStart = 0;
75     digitalWrite(greenLED, HIGH);
76     if (lastState != "RESET") {
77       setLCD(" RESET ", " System SAFE ");
78       lastState = "RESET";
79     }
80     Serial.println("RESET");
81     delay(500);
82     return;
83   }
84
85   // --- FIRE / EVACUATE ---
86   if (gasVal > 800 && tempC > 60) {
87     if (!timerOn) {
88       timerOn = true;
89       alarmStart = millis();
90     }
91
92     if (millis() - alarmStart > 10000) {
93       alloff();
94       if (lastState != "EVACUATE") {
95         setLCD("!!! FIRE !!! ", "EVACUATE NOW!! ");
96         lastState = "EVACUATE";
97         Serial.println("EVACUATE");
98       }
99       digitalWrite(buzzer, HIGH);
100      digitalWrite(redLED, HIGH);
101      delay(200);
102      digitalWrite(redLED, LOW);
103      delay(200);
104      return;
105    }
106  }
```

```
107 alloff();
108 if (lastState != "ALARM") {
109   setLCD(" !! ALARM !! ", " Temp+Gas High ");
110   lastState = "ALARM";
111   Serial.println("ALARM");
112 }
113 digitalWrite(redLED, HIGH);
114 digitalWrite(buzzer, HIGH);
115 delay(120);
116 digitalWrite(buzzer, LOW);
117 delay(120);
118 return;
119 }
120
121 timerOn = false;
122 alarmStart = 0;
123
124 // --- WARNING ---
125 if (gasVal > 700 || tempC > 35) {
126   alloff();
127   if (lastState != "WARNING") {
128     setLCD(" !! WARNING !! ", " Check Area ");
129     lastState = "WARNING";
130     Serial.println("WARNING");
131   }
132   digitalWrite(yellowLED, HIGH);
133   digitalWrite(buzzer, HIGH);
134   delay(600);
135   digitalWrite(buzzer, LOW);
136   delay(600);
137   return;
138 }
139
140 // --- SAFE ---
141 alloff();
142 digitalWrite(greenLED, HIGH);
143
144 if (lastState != "SAFE") {
145   setLCD(" SAFE : ) ", "Temp: ");
146   lastState = "SAFE";
147 }
148 // Live temp update on line 2
149 lcd.setCursor(5, 1);
150 lcd.print(tempC, 1);
151 lcd.print((char)223); // ° symbol
152 lcd.print("C ");
153
154 Serial.println("SAFE");
155 delay(500);
156 }
```

Serial Monitor

```
SAFE
Gas: 487 Temp: 24.78
RESET
Gas: 487 Temp: 24.78
SAFE
Gas: 487 Temp: 24.78
SAFE
```

TINKERCAD SERIAL MONITOR VIEW



TECHNICAL DEMO

Demo Platform: Tinkercad Circuits (online simulation)

What the demo shows:

- ✓ Normal state: Sensors reading low values → No alarm, LED off, buzzer silent
- ✓ Trigger state: Smoke level exceeds threshold → Buzzer activates, Red LED turns ON
- ✓ Code behaviour: Arduino reads analog input from MQ-2 → if value > 300, set buzzer and LED to HIGH

Component roles in demo:

MQ-2: Provides analog voltage signal proportional to smoke concentration

Arduino Uno: Processes signal and executes alarm logic

Buzzer: Produces audible alarm output

Red LED: Provides visible alert confirmation

Visual tip: Take a clear screenshot of your Tinkercad circuit in both OFF and ON states – place them side by side on the slide

Demo Reproduce Notes:

1. Open the SimulationOpen the Tinkercad simulation link provided in this document and click Start Simulation to run the system.
2. Hardware Components UsedThe system consists of the following components:Arduino UNOTemperature sensor (TMP)Push button (manual trigger)LEDs (Green, Yellow, Red)BuzzerLCD DisplayResistorsBreadboard and jumper wires
3. System InputsThe system uses three logical inputs:S – Sensor input (temperature condition)T – Trigger input (manual button press) R – Reset/Safe condition
4. Warning ConditionThe Warning signal is generated using the logic: $W = S \text{ OR } T$ If either the sensor detects a condition or the manual trigger is pressed, the warning LED turns ON.
5. Alarm ConditionThe Alarm signal is generated using the logic: $A = S \text{ AND } T$ When both the sensor detects danger and the manual trigger is activated, the alarm LED and buzzer activate.
6. Evacuation ConditionThe Evacuation signal is generated using the logic: $E = (S \text{ AND } T) \text{ AND NOT } R$ If the danger condition exists and the reset signal is OFF, the evacuation alert is triggered.
7. Output IndicatorsThe system provides outputs using:LED indicatorsBuzzer alertLCD display messagesThe LCD shows the current system status such as Warning, Alarm, or Evacuate.
8. Testing the SystemTo reproduce the demo:Start the simulation in Tinkercad.Change the sensor value or press the push button.Observe the LED indicators and buzzer activation.The LCD display updates according to the logic conditions.This confirms the correct operation of the logic gate design and Arduino control system.